SAE International™

SURFACE VEHICLE RECOMMENDED PRACTICE

SAE J1939-82 AUG2008

Issued 2008-08

Compliance—Truck and Bus

RATIONALE

Not applicable.

FOREWORD

This series of SAE Recommended Practices has been developed by the Truck and Bus Control and Communications Network Subcommittee of the Truck and Bus Electrical and Electronics Committee. The objectives of the subcommittee are to develop information reports, recommended practices, and standards concerned with the requirements, design, and usage of devices that transmit electronic signals and control information among vehicle components. The usage of these Recommended Practices is not limited to truck and bus applications. Other applications may be accommodated with immediate support being provided for construction and agricultural equipment, and stationary power systems. These SAE Recommended Practices are intended as a guide toward standard practice and are subject to change so as to keep pace with experience and technical advances.

This particular document, SAE J1939-82, describes the compliance tests and procedures to verify an SAE J1939 ECU will operate correctly on a SAE J1939 network. This compliance document is used for all SAE J1939 applications.

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1. SCOPE

The purpose of these compliance procedures is to generate one or more test documents that outline the tests needed to assure that an ECU that is designed to operate as a node on a SAE J1939 network would do so correctly. SAE does not certify devices and these tests and their results do not constitute endorsement by SAE of any particular device. These tests are presented to allow testing of a Device to determine self-compliance by the manufacturer of a device. The manufacturer can use its record of what procedures were run successfully to show the level of compliance with SAE J1939.

2. REFERENCES

This specification takes precedence over all conflicts in the documents cited in this section.

2.1 Applicable Publications

The following publications form a part of this specification to the extent specified herein. Unless otherwise specified, the latest issue of SAE publications shall apply.

2.1.1 SAE Publications

Available from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, Tel: 877-606-7323 (inside USA and Canada) or 724-776-4970 (outside USA), www.sae.org.

SAE J1455 Recommended Environmental Practices for Electronic Equipment Design in Heavy-Duty Vehicle Applications

SAE J1939 Recommended Practice for a Serial Control and Communications Vehicle Network

SAE J1939-11 Physical Layer, 250K bits/s, Twisted Shielded Pair

SAE J1939-15 Reduced Physical Layer, 250K Bits/Sec, Un-Shielded Twisted Pair (UTP)

SAE J1939-21 Data Link Layer

SAE J1939-31 Network Layer

SAE J1939-81 Network Management

2.1.2 ISO Publications

Available from American National Standards Institute (ANSI), 25 West 43rd Street, New York, NY 10036-8002, Tel: 212-642-4900, www.ansi.org.

ISO 11898-1 Road vehicles—Controller area network (CAN)—Part 1: Data link layer and physical signaling

3. DEFINITIONS

Terms and definitions are defined in SAE J1939, except the following:

3.1 Certification

Public announcement, using documentation with signatures from a member of a duly recognized competent body (i.e. UL, TUV, CSA), to give notice that a given device has been tested and found to meet all necessary issues of a particular requirement or standard, whether legislated or of purely common industrial usage.

3.2 Compliance

Announcement that a device has been tested and found to meet a particular set (not necessarily all) of issues of a particular requirement or standard, without any supporting signatures from a recognized standard agency as performed by a third party investigator.

3.3 Self-Compliance

Announcement that a device has been tested and found to meet a particular set (not necessarily all) of issues of a particular requirement or standard, without any supporting signatures from a recognized standard agency.

4. ABBREVIATIONS

CTC Compliance Test Computer
DUT Device Under Test

5. EQUIPMENT

The Standard Test Configuration will contain two nodes, the Compliance Test Computer (CTC) and the Device Under Test (DUT), with appropriate media and termination, as well as power supplies and DUT loads.

5.1 Compliance Test Computer (CTC)

A Compliance Test Computer with the following capabilities:

5.1.1 Message Transmission

Must be able to send any specified message.

5.1.2 Message Reception

Must be able to receive all bus messages.

5.1.3 Time Stamp

Must be able to time stamp at the beginning of received messages at the required resolution.

5.1.4 Time Resolution

Must possess a minimum time resolution of 10 microsecond. The actual time resolution must be specified.

5.1.5 Time Stamp Accuracy

The error between any two timestamps shall be less than 1% of the time difference between the two timestamps or 1 millisecond, whichever is greater.

5.1.6 Time Stamp Latency Error

Must be less than 1 count of the time resolution.

5.1.7 Transmission Synchronization

Must be able to control the time between messages it transmits.

5.1.8 Reception Timeout

In cases where DUT should have ‘no response’ the CTC must be able to monitor that no response message is transmitted by the DUT for a minimum time equal to the allowed transmission response time plus two times the allowed latency (for example when transmission should be within 200 milliseconds and 50 millisecond latency is allowed then CTC must check for a minimum time of 300 milliseconds). Longer times are at the discretion of the designers.

5.1.9 CAN Interface

At least one CAN interface is required. Two CAN interfaces are required for testing Bridge functions.

5.2 Device Under Test (DUT)

The Device Under Test with the following features:

5.2.1 Operation

Must be able to perform its normal intended use (not operating any special testing mode of operation).

5.2.2 Controls

Must be able to initiate the action under test when directed.

6. SETUP

The setup will consist of a minimum network with two nodes. Any additional requirements for a specific test procedure will be specified within that procedure.

6.1 Minimum Network

Two nodes, the DUT and the CTC, with terminations, a short length of the appropriate media, power supplies.

6.2 Message Traffic

Minimal normal network traffic, only the messages necessary for the specific test procedure (see 5.2.1).

6.3 Test Message Set

Messages that are applicable to a specific test procedure will be listed in the test outline. This message list will be broken into two pieces: those supported for reception and those supported for transmission.

7. TEST SELECTION

The procedure to select a test or set thereof, necessary to check for compliance to a particular function or set of functions from the standard, are outlined within Appendix A. Appendix A is broken into tables, which identify functions, based on which document section(s) they are described within. Some tests will be pointed to by more than one entry.

7.1 Document Identifying Test

The specific document identifying a particular function will be outlined with any necessary cross-references within a Table for the specific task force document.

7.2 Requirement versus Device Characteristics

The ‘SAE and User Requirements’ status of each function will be cross referenced with the Device Characteristics (such as: Arbitrary Address Capable, Command Configurable, Diagnostic Tool, etc.) that will affect whether a device should provide said function.

8. RUNNING A TEST

The steps necessary to run a test are outlined as follows:

8.1 Identify Test

Using Appendix A the reviewer should identify the test number(s) necessary to check any particular function(s) that they desire. Note that all tests are not applicable to all systems. The “Requiring Document” column refers the reviewer to the appropriate document section the test is intended to cover.

8.2 Report Generation

Using APPENDIX A the reviewer should document the test(s) performed and the results, as well as whether the device had requirement exceptions (additions or exclusions) for each of the particular functions.

9. REPORTING COMPLIANCE TEST RESULTS

Appendix A, Compliance Tests, contains columns in the forms for reporting the results of compliance testing. These columns are used to report the test date, whether a particular test is required for the specific device being tested and whether the test has been run, as well as whether the device passed (complied with) the test.

9.1 Self-Compliance

Appendix A is intended to be used by an OEM as self-compliance documentation. When certifying a device the OEM can indicate whether that the device is intended to provide each function, as well as the general level to which the device is designed. The passage of each function by the particular device can then be documented along with the date of testing. The inclusion of the specific test number(s) should enable future comparison among devices, as well as, among testers.

9.2 Device Support

The reporting forms record the test(s) conducted, the test results, and whether a device is intended to support a particular function, which makes these reporting forms very helpful to a customer locating a desired device.

PREPARED BY THE SAE TRUCK AND BUS CONTROL AND COMMUNICATIONS NETWORK SUBCOMMITTEE
OF THE SAE TRUCK AND BUS ELECTRICAL AND ELECTRONICS COMMITTEE

APPENDIX A - COMPLIANCE TESTS

A.1 INTRODUCTION

Tables herein describe tests and/or procedures needed to check compliance of a device against each requirement of the SAE J1939 document set. The sections are divided by document title with section number references to the specific text defining a function and its operation. The tables include all described network functions not simply minimum requirements. The requirement to perform a test or procedure for an ECU can be discerned by looking in the applicable ECU Class columns under the ‘SAE and User Requirements’ heading. See Section A.2.5 below for more details about interpreting testing requirements.

A.2 TEST OUTLINE TABLE STRUCTURE AND COLUMN DEFINITIONS

The following definitions and values are identified to enable completion of the entries into the respective columns of the compliance test tables.

A.2.1 Row

Provides a numeric tag to use in references to particular tests.

A.2.2 Test Name

Name or title for the particular test.

A.2.3 Requiring Document

Specifies the source of the particular item. Usually this will be a particular section of a SAE J1939-xx document.

A.2.4 Description

Provides a short outline of the identified feature and the test.

A.2.5 SAE and User Requirements

The SAE and User Requirements consists of several ‘ECU Classes’ columns and a User column. The set of columns under the 'ECU Class' heading are used to identify the necessity to check compliance to a particular test, based upon the ECU class. The column under the 'User Add./Excl.' heading is provided to allow users to customize the necessity to check compliance to a particular test. An ‘X’ in any of the ‘ECU Classes’ columns implies the test is required by SAE for that ECU Class (see A.2.5.2 below for ECU Class definition). A ‘D’ in the User column implies this test is desired as an additional feature for the particular ECU under test. An ‘E’ implies that the ECU under test is not required to provide this feature even when it might be a requirement.

- The presence of a code in an 'ECU Class' column conveys a requirement to evaluate the test for an ECU of that class.
- The presence of a code in the 'User Add./Excl.' column conveys a requirement to evaluate the test per the requirements of the 'user'.
- The absence of a code in a 'SAE and User Requirements' column indicates there is no explicit requirement to evaluate the test for an ECU of that class or per the requirements of the 'user'.

A.2.5.1 Requirement Codes

There are currently three (3) codes defined to denote the requirement for evaluating a particular test. New codes will be added as the need is identified. The test requirement codes supported for the Test Outlines are summarized in Table A1.

TABLE A1 - TEST REQUIREMENT CODES

Code	Meaning	Specified By
X	Compliance to the test is Required by SAE for that ECU Class	SAE J1939 Subcommittee
D	Compliance to the test is Desired as an additional feature for the particular ECU under test	User
E	Compliance to the test is not required (i.e. may be excluded) for the test ECU, even though the test might be a requirement for the ECU Class	User

A.2.5.2 For example, an ‘X’ in any of the ‘ECU Classes’ columns implies the test is required by SAE for that ECU Class. A ‘D’ in the 'User Add./Excl.' column implies this test is desired as an additional feature for the particular ECU under test. An ‘E’ in the 'User Add./Excl.' column implies that the ECU under test is not required to provide this feature even when it might be a requirement. The marking of these extra conditions would enable the tables to be used for test reporting or purchasing specification, as well as to identify all the compliance tests. ECU Class

There are currently seven (7) ECU Classes defined for testing requirement association. New ECU Classes will be added as the need is identified. The ECU Class codes supported for the Test Outlines are summarized in Table A2.

TABLE A2 - ECU CLASS CODES

ECU Class Code	Code Description	Description of Applicable ECUs
ALL	All ECUs	All SAE J1939 ECUs, regardless of the ECU’s design function or specifications
BDG	Network Interconnect ECUs	General network interconnect devices (bridge, router, etc…)
AAC	'Arbitrary Address Capable' ECUs	ECUs or Controller Applications that are ‘Arbitrary Address Capable’ regarding address claim
CC	'Command Configurable' Address ECUs	ECUs or Controller Applications that support ‘Command Configurable’ addressing
SC	'Self Configurable' Address ECUs	ECUs or Controller Applications that are ‘Self Configurable’ regarding address claim
TOO	Tool	ECU’s that are to function as a ‘tool’
EMS	On-Highway Emissions Regulated ECUs	ECU’s are operating with On-Highway Emissions regulated engines

A.2.5.3 User Add./Excl.

The 'User Add./Excl.' column provides a place for the "User" to indicate the Addition (D) or Exclusion (E) of the test for a test ECU. The “User” is the user of this document and could be a purchasing agent, a system specifier, a supplier, a tester, etc. The marking of these extra conditions enable the tables to be used for test reporting or other specification, such as purchasing specification, as well as to identify all the compliance tests.

A.2.6 Date Tested

Identifies the date the test was conducted.

A.2.7 Pass - Fail

Record the test results. The designation choices are

P for Pass
F for Fail

A.2.8 Test Result Comments

Record the comments the tester or others may choose to include.

A.3 COMPLIANCE TESTS FOR DATA LINK LAYER (J1939-21)

A.3.1 Data Link Layer Tests - Transmit Behavior

The Compliance Tests in Table A3 evaluate the general message transmit behaviors of the DUT. The tests for evaluating the behaviors of the DUT as the data source using the Transport Protocols are presented in Table A5 and Table A7.

TABLE A3 - DATA LINK TRANSMIT TESTS - GENERAL

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded — ECU Classes: ALL	BDG	AAC	CC	SC	TOO	EMS	User Add./ Excl	Date Tested	Pass-Fail	Test Result Comments
1	Position and order of PGN in the 29-bit Frame Header (DUT as Source)	J1939-21 5.1.3 Table 1	Verify proper placement of PGN within 29-bit header of all frames	X										
2	Use of EDP Bit to identify PGN (DUT as Source)	J1939-21 5.2.2	Verify DUT properly sets the EDP Bit for all 29-bit frames.	X										
3	Use of DP Bit to identify PGN (DUT as Source)	J1939-21 5.2.3	Verify DUT properly sets the DP Bit for all 29-bit frames.	X										
4	Data Field Byte Length (DUT as Source)	J1939-21 5.2.7.1	Verify the length of the data field for all 29-bit Frames (Page 1 & 2) is 8 bytes, except for PGN 59904 frames.	X										
5	Byte Ordering within Data Field (DUT as Source)	J1939-21 5.4	Verify proper byte ordering of the parameter data for multiple byte parameters. (e.g. Verify Address Claim data, PGN 60928). This verification also applies to -7X.	X										
6	Multipacket capable PGN with less than 9 bytes of data sent as single CAN Data Frame with 8 byte data field. (DUT as Source)	J1939-21 5.10.5.2 5.2.7.1 5.2.7.2	Verify DUT indeed uses single CAN Frame to send PGN under this situation. Verify any unused bytes are filled with $FF.	X										

TABLE A3 - DATA LINK TRANSMIT TESTS - GENERAL (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
7	Request Scheduling (DUT as Source)	J1939-21 5.12.2	Verify the DUT does not send a Request message for a PGN if that PGN was received with the last 50 mS	X										
8	Request Retries (DUT as Source)	J1939-21 5.12.3	Verify the DUT stops sending the same Request message after the third attempt (second retry). A Request retry is issued following a Response Timeout (Tr) failure.	X										
9	Correct Interpretation of 'Requested PGN' in Request (PGN 59904) (DUT as Source)	J1939-21 5.4.2 Table 6	Verify Correct Request message structure Verify 'Requested PGN' in Request sent by DUT has correct content (order and position)	X										
10	Response Timing (DUT as Source)	J1939-21 5.12.3	Verify DUT waits 1.25 S (T3) for a required response before retrying or quitting	X										
11	Devices Responds to its own Global Read Requests (DUT as Source)	J1939-21 5.12.4	Verify DUT sends a response to its own Global Read request within the Tr Time Verify the DUT uses the appropriate type of response	X										
12	Support of the Acknowledgement Address data byte for each ACK and NACK (DUT as Source)	J1939-21 5.4.4	Verify DUT evaluates the 'Acknowledgement Address' data field byte to determine the context of the Acknowledgement (ACK and NACK of same PGN with different 'Acknowledgement Address' data values)	X										

A.3.2 Data Link Layer Tests - Receive Behavior

The Compliance Tests in Table A4 evaluate the general message receive behaviors of the DUT. The tests for evaluating the behaviors of the DUT as the data recipient using Transport Protocol are presented in Table A6 and Table A8.

TABLE A4 - DATA LINK RECEIVE TESTS - GENERAL

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
1	Standard Frame Message Tolerance	J1939-21 5.1.3	Verify DUT not affected by standard frames											
2	Device not a CAN 2.0A Device	J1939-21 5.1.3	Verify DUT not CAN 2.0A device by issuing 29-bit (CAN 2.0B) frames											
3	Independence of Priority Bits in PGN Receive (DUT as Recipient)	J1939-21 5.2.1	Verify DUT receives a PGN regardless of the priority bits in 29-bit header. Change priority bits and confirm received.	X										
4	Use of EDP Bit to identify PGN (DUT as Recipient)	J1939-21 5.2.2	Verify DUT evaluates the EDP Bit when processing in 29-bit frames. (Same CAN header except for EDP bit and different data values to see if DUT acts on data values)	X										
5	Use of DP Bit to identify PGN (DUT as Recipient)	J1939-21 5.2.3	Verify DUT evaluates the DP Bit when processing in 29-bit frames.	X										
6	Filtering on Destination Address (DUT as Recipient)	J1939-21 5.2.5.1	Verify DUT evaluates the Destination Address for 29-bit frames. (Same PGN to different DA with different data values to see if DUT acts on data values)	X										
7	Supports Receive of Global Destination Address (DUT as Recipient)	J1939-21 5.2.5.1	Verify DUT responds to globally addressed messages.	X										

TABLE A4 - DATA LINK RECEIVE TESTS - GENERAL (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
8	PDU Processing Capabilities (DUT as Recipient)	J1939-21 5.11	Verify device does not lose messages when the data link is at 100 percent utilization for 10 mS. Verify device does not lose back-to-back messages when the data link is at 100 percent utilization for 10 mS	X										
9	Multipacket capable PGN with less than 9 bytes of data sent as single CAN Data Frame with 8 byte data field. (DUT as Recipient)	J1939-21 5.10.5.2 5.2.7.1 5.2.7.2	Verify DUT receives the PGN in the single CAN Frame format.	X										
10	Concurrent receive of Multipacket capable Destination Specific (PDU1) PGN - one instance with less than 9 bytes (single CAN Frame) and the other instance with more than 9 bytes (RTS/CTS) (DUT as Recipient)	J1939-21 5.10.5.3 5.2.7.1 5.2.7.2	Verify DUT can receive both concurrent instances of same PGN	X										
11	Concurrent receive of Multipacket capable Broadcast (PDU2) PGN - one instance with less than 9 bytes (single CAN Frame) and the other instance with more than 9 bytes (BAM) (DUT as Recipient)	J1939-21 5.10.5.3 5.2.7.1 5.2.7.2	Verify DUT can receive both concurrent instances of same PGN	X										
12	Correct Interpretation of 'Requested PGN' in Request (PGN 59904) (DUT as Recipient)	J1939-21 5.4.2 Table 6	Verify DUT properly interprets 'Requested PGN' in Request message by monitoring for correct PGN response	X										

TABLE A4 - DATA LINK RECEIVE TESTS - GENERAL (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
13	Proper Response to Destination Specific Request for Single Packet Destination Specific (PDU1) PGN (DUT as Recipient)	J1939-21 5.4.2 Table 5	Verify DUT sends the Requested PGN with the Destination Address set to the Source Address from the Request message Verify DUT sends response within 200 mS (Tr) after the Request	X										
14	Proper Response to Global Request for Single Packet Destination Specific (PDU1) PGN (DUT as Recipient)	J1939-21 5.4.2 Table 5	Verify DUT sends the Requested PGN with the Destination Address set to the Global Address Verify DUT sends response within 200 mS (Tr) after the Request	X										
15	Proper Response to Destination Specific Request for Multipacket Destination Specific (PDU1) PGN (DUT as Recipient)	J1939-21 5.4.2 Table 5	Verify DUT sends a SAE J1939 Transport RTS for the Requested PGN to the Source Address from the Request message Verify DUT sends the RTS response within 200 mS (Tr) after the Request	X										
16	Proper Response to Global Request for Multipacket Destination Specific (PDU1) PGN (DUT as Recipient)	J1939-21 5.4.2 Table 5	Verify DUT sends a SAE J1939 Transport BAM for the Requested PGN Verify DUT sends the BAM response within 200 mS (Tr) after the Request											
17	Proper Response to Destination Specific Request for Single Packet Broadcast (PDU2) PGN (DUT as Recipient)	J1939-21 5.4.2 Table 5	Verify DUT sends the Requested PGN Verify DUT sends response within 200 mS (Tr) after the Request	X										
18	Proper Response to Global Request for Single Packet Broadcast (PDU2) PGN (DUT as Recipient)	J1939-21 5.4.2 Table 5	Verify DUT sends the Requested PGN Verify DUT sends response within 200 mS (Tr) after the Request	X										

TABLE A4 - DATA LINK RECEIVE TESTS - GENERAL (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
19	Proper Response to Destination Specific Request for Multipacket Broadcast (PDU2) PGN (DUT as Recipient)	J1939-21 5.4.2 Table 5	Verify DUT sends a SAE J1939 Transport RTS for the Requested PGN to the Source Address from the Request message Verify DUT sends the RTS response within 200 mS (Tr) after the Request	X										
20	Proper Response to Global Request for Multipacket Broadcast (PDU2) PGN (DUT as Recipient)	J1939-21 5.4.2 Table 5	Verify DUT sends a SAE J1939 Transport BAM for the Requested PGN Verify DUT sends the BAM response within 200 mS (Tr) after the Request	X										
21	Response Timing (DUT as Recipient)	J1939-21 5.12.3	Verify DUT sends all required responses within 200 mS (Tr)	X										
22	Proper NACK Response for Destination Specific Request for Unsupported PGN (DUT as Recipient)	J1939-21 5.4.4, 5.4.2 Table 5	Verify DUT does nothing if it wasn't the Destination of the Request Verify DUT uses the Global Address for the message Verify DUT sends NACK within 200 mS (Tr)	X										
23	Proper NACK Response for Globally Addressed Request for Unsupported PGN (DUT as Recipient)	J1939-21 5.4.4, 5.4.2 Table 5	Verify DUT does not send any Acknowledgement message PGN 59392) Monitor for DUT messages for 1.25 S (T3) to verify the DUT does not send an Acknowledgement for the requested PGN	X										

TABLE A4 - DATA LINK RECEIVE TESTS - GENERAL (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
24	Proper use of ACK Response when Applicable (DUT as Recipient)	J1939-21 5.4.4, 5.4.2 Table 5	Verify DUT sends an Acknowledgement (PGN 59392) with Requested PGN and Control Byte of ACK (= 0) Verify DUT sends ACK only if it was the Destination of the Request Verify DUT does nothing if it wasn't the Destination of the Request Verify DUT uses the Global Address for the message Verify DUT sends ACK within 200 mS (Tr)	**X**										
25	Proper use of Access Denied NACK Response (DUT as Recipient)	J1939-21 5.4.4, 5.4.2 Table 5	Verify DUT sends an Acknowledgement (PGN 59392) with Requested PGN and Control Byte of NACK Access Denied (= 2) Verify DUT sends NACK only if it was the Destination of the Request Verify DUT does nothing if it wasn't the Destination of the Request Verify DUT uses the Global Address for the message Verify DUT sends ACK within 200 mS (Tr)	**X**										

TABLE A4 - DATA LINK RECEIVE TESTS - GENERAL (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
26	Proper use of Cannot Respond NACK Response (DUT as Recipient)	J1939-21 5.4.4, 5.4.2 Table 5	Verify DUT sends an Acknowledgement (PGN 59392) with Requested PGN and Control Byte of NACK Cannot Respond (= 3) Verify DUT sends NACK only if it was the Destination of the Request Verify DUT does nothing if it wasn't the Destination of the Request Verify DUT uses the Global Address for the message Verify DUT sends ACK within 200 mS (Tr)	X										
27	Support of the Acknowledgement Address data byte for each ACK and NACK (DUT as Recipient)	J1939-21 5.4.4	Verify the 'Acknowledgement Address' data field byte contains a valid or appropriate address value	X										
28	Correct Response to Request2 with 'yes' for the "Use Transfer PGN" Option (DUT as Recipient)	J1939-21 5.4.6 5.4.7	Verify the response data field byte contains a valid or appropriate address value	X										
29	Correct Response to Request2 with 'no' for the "Use Transfer PGN" Option (DUT as Recipient)	J1939-21 5.4.6	Verify the response data field byte contains a valid or appropriate address value	X										
30	Proper NACK Response for Request2 for Unsupported PGN (DUT as Recipient)	J1939-21 5.4.6	Verify the response data field byte contains a valid or appropriate address value	X										

A.3.3 Data Link Layer Tests - BAM Originator Behaviors

The Compliance Tests in Table A5 evaluate the behaviors of the DUT as the data originator for the SAE J1939 Transport Protocol BAM service. The SAE J1939 Transport Protocol RTS/CTS tests for the data originator are presented in Table A7.

TABLE A5 - DATA LINK TRANSMIT TEST - TP BAM

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
1	BAM Protocol: BAM is valid (content and format) (DUT as Originator)	J1939-21 5.10.2.1 5.10.3 5.10.3.5	Verify correct PGN, data size, & # packets. Verify all match the BAM TP.DT from the DUT.	X										
2	BAM Protocol: BAM is sent before Data Packets (DUT as Originator)	J1939-21 5.10.2.1 5.10.1.3	Verify DUT sends BAM prior to Data Packets Verify time between BAM and first Data Packet is is between 50 and 200 mS (5.10.1.3)	X										
3	BAM Protocol: Verify no Conn_Abort is sent (DUT as Originator)	J1939-21 5.10.2.4	Verify DUT does not send a TP.Conn_Abort for any reason for a BAM Verify DUT ignores a TP.Conn_Abort received for the PGN of the BAM	X										
4	BAM Protocol: Only one per Originator at a time (DUT as Originator)	J1939-21 5.10.5.1	Verify DUT doesn't start a BAM a previous BAM data transferred	X										
5	BAM Protocol: Simultaneous BAMs with different Originators (DUT as Originator)	J1939-21 5.10.5.1	Verify DUT doesn't start a BAM a previous BAM data transferred	X										
6	Transport Data Packets content correct (DUT as Originator)	J1939-21 5.10.1.3 5.10.2.3 5.10.4	Verify first data byte is sequence number Verify remaining 7 bytes are correct data for packet	X										

TABLE A5 - DATA LINK TRANSMIT TEST - TP BAM (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
7	BAM Transport Data Packets sent in ascending sequential order (DUT as Originator)	J1939-21 5.10.1.1 5.10.1.2 5.10.1.4	Verify DUT sends the TP Data Packets for BAM in sequential order Verify the first TP.DT packet is sequence number = 1	X										
8	All Transport Data packets (PGN 60160) have an 8 byte data field (DUT as Originator)	J1939-21 5.10.1.3 5.2.7.2 5.10.4	Verify DUT sends packets with 8 bytes	X										
9	Unused bytes of Last Transport Data packets (PGN 60160) filled with "FF_{16}" (DUT as Originator)	J1939-21 5.10.1.3 5.10.4	Verify unused bytes of last packets sent & filled properly	X										
10	Time between Transport Data Packets for BAM (DUT as Originator)	J1939-21 5.10.1.3	Verify packets are sent between 50 and 200 mS apart	X										

A.3.4 Data Link Layer Tests - BAM Receive Behaviors

The Compliance Tests in Table A6 evaluate the behaviors of the DUT as the data recipient for the SAE J1939 Transport Protocol BAM service. The SAE J1939 Transport Protocol RTS/CTS tests for the data recipient are presented in Table A8.

TABLE A6 - DATA LINK RECEIVE TESTS - TP BAM

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
1	BAM Protocol: BAM is valid (content and format) (DUT as Responder)	J1939-21 5.10.2.1 5.10.3 5.10.3.5	Verify behavior when receiving the BAM with correct PGN placed, data size, & Number of Data packets Verify behavior when receiving the BAM with incorrect PGN placed, data size, & Number of Data packets	X										
2	BAM Protocol: BAM is sent before Data Packets (DUT as Responder)	J1939-21 5.10.2.1 5.10.1.3	Verify behavior when time between BAM and first packet is between 50 and 200 mS Verify behavior when time between BAM and first packet is faster than 50 mS Verify behavior when time between BAM and first packet is longer than 200 mS	X										
3	BAM Protocol: Verify no EndOfMsgACK is sent after final Data Packet (DUT as Responder)	J1939-21 5.10.2.4	Verify DUT does not sent an TP.CM_EndOfMsgACK after the final data packet of the BAM	X										
4	BAM Protocol: Verify no Conn_Abort is sent (DUT as Responder)	J1939-21 5.10.2.4	Verify DUT does not sent an TP.Conn_Abort for any reason for a BAM	X										
5	BAM Protocol: Only one per Originator at a time (DUT as Responder)	J1939-21 5.10.5.1	Verify behavior when sees a BAM from an originator before previous BAM data transferred	X										

TABLE A6 - DATA LINK RECEIVE TESTS - TP BAM (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
6	BAM Protocol: Simultaneous BAMs with different Originators (DUT as Responder)	J1939-21 5.10.5.1	Verify ability of DUT to successfully receive multiple simultaneous BAMs from different originators	X										
7	BAM Transport Data Packets sent in ascending sequential order (DUT as Responder)	J1939-21 5.10.1.1 5.10.1.2 5.10.1.4	Verify DUT behavior when the TP Data packets for a BAM are received in sequential order Verify behavior when TP Data packets for a BAM are received out of sequential order	X										
8	BAM Data Packets: Time between Transport Data Packets for BAM (DUT as Responder)	J1939-21 5.10.1.3	Verify behavior when packets are sent between 50 and 200 mS apart Verify behavior when packets are sent faster than 50 mS apart Verify behavior when packets are sent longer than 200 mS apart	X										

A.3.5 Data Link Layer Tests - RTS/CTS Transport Protocol Originator Behaviors

The Compliance Tests in Table A7 evaluate the behaviors of the DUT as the data originator for the SAE J1939 Transport Protocol RTS/CTS service. The SAE J1939 Transport Protocol BAM tests for the data originator are presented in Table A5.

TABLE A7 - DATA LINK TRANSMIT TESTS - TP RTS/CTS

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
1	RTS/CTS Protocol: RTS is sent before starting Data Transfer (DUT as Originator)	J1939-21 5.10.2.2	Verify DUT sends RTS to initiate connection	X										
2	RTS/CTS Protocol: RTS is valid (content and format) (DUT as Originator)	J1939-21 5.10.2.2 5.10.3 5.10.3.1	Verify correct PGN, data size, & total # packets, max packets. Verify all match the RTS/CTS TP.DT from the DUT	X										
3	RTS/CTS Protocol: CTS response to RTS is valid (content and format) (DUT as Originator)	J1939-21 5.10.2.2 5.10.3 5.10.3.2	Verify behavior when CTS sent within 200 mS (Tr) following RTS Verify behavior when CTS sent after more than 200 mS (Tr) following RTS Verify DUT checks PGN Verify behavior when # packets (CTS) > max packets (RTS) Verify behavior for wrong/invalid next packet	X										
4	RTS/CTS Protocol: To reject, RTS is followed by valid Conn_Abort (content and format) (DUT as Originator)	J1939-21 5.10.2.2 5.10.3.4	Verify DUT terminates or stops in response to reject (Abort) Verify DUT checks PGN	X										

TABLE A7 - DATA LINK TRANSMIT TESTS - TP RTS/CTS (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
5	RTS/CTS Protocol: CTS issued after last Transport Data packet of previous CTS (DUT as Originator)	J1939-21 5.10.2.2 5.10.2.4 5.10.3 5.10.3.2	Verify behavior when CTS sent within T3 time following last packet Verify behavior when CTS sent after more than T3 time following last packet Verify DUT checks PGN Verify behavior when # packets (CTS) > max packets (RTS) Verify behavior for wrong/invalid next packet	X										
6	RTS/CTS Protocol: CTS next packet numbering (DUT as Originator)	J1939-21 5.10.3.2	Verify behavior if next packet number specified in CTS is same as next packet number of previous CTS Verify behavior if next packet number specified in CTS is within packet range of previous CTS Verify behavior if next packet number specified in CTS is less than last confirmed packet number Verify behavior if next packet number specified in CTS is one greater than next packet number plus number of packets from last CTS Verify behavior if next packet number specified in CTS is two or more greater than next packet number plus number of packets from last CTS	X										

TABLE A7 - DATA LINK TRANSMIT TESTS - TP RTS/CTS (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
7	RTS/CTS Protocol: CTS to pause or stop data flow (DUT as Originator)	J1939-21 5.10.2.3 5.10.3 5.10.3.2	Verify behavior when receives CTS to pause Verify DUT checks PGN Verify behavior when CTS sent within T3 time following last packet Verify behavior when CTS sent after more than T3 time following last packet	X										
8	RTS/CTS Protocol: CTS to pause repeat (DUT as Originator)	J1939-21 5.10.2.3 5.10.2.4 5.10.3 5.10.3.2	Verify behavior when receives CTS to pause within 500 mS (Th) Verify behavior when receives CTS to pause exceeds 500 mS (Th) Verify behavior when receives CTS to pause exceeds T4 Time Verify DUT checks PGN	X										
9	RTS/CTS Protocol: EndOfMsgACK (DUT as Originator)	J1939-21 5.10.2.4 5.10.3 5.10.3.3	Verify behavior when receives TP.CM_ EndOfMsgACK within T3 Time Verify behavior when receives TP.CM_ EndOfMsgACK after more than T3 Time Verify ignores TP.CM_ EndOfMsgACK if received before final Data Transfer Verify DUT checks PGN	X										
10	RTS/CTS Protocol: Connect Abort issued by Originator (DUT as Originator)	J1939-21 5.10.2.4 5.10.3.4	Verify correct content in TP.Conn_Abort Verify only sends once connection is established (i.e. after initial Accept CTS)	X										

TABLE A7 - DATA LINK TRANSMIT TESTS - TP RTS/CTS (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
11	RTS/CTS Protocol: Connect Abort issued by Responder (DUT as Originator)	J1939-21 5.10.2.4 5.10.3.4	Verify behavior when receives TP.Conn_Abort Verify DUT checks PGN Verify TP Data stops within 50 mS after TP.Conn_Abort Verify TP Data sends no more than 32 packets after TP.Conn_Abort	X										
12	RTS/CTS Protocol: Multiple duplicate CTS (same SA, DA, PGN) received in short period (DUT as Originator)	J1939-21 5.10.3.2	Verify behavior (abort) with received multiple duplicate CTS (excluding CTS(0))	X										
13	RTS/CTS Protocol: CTS outside of connection is ignored (DUT as Originator)	J1939-21 5.10.3.2	Verify ignores CTS for a PGN that is not part of a connection	X										
14	RTS/CTS Protocol: Another RTS with same Originator and Responder but with different PGN (DUT as Originator)	J1939-21 5.10.5	Verify DUT does not send an RTS for another PGN to Responder while another RTS/CTS open with same Responder	X										
15	RTS/CTS Protocol: Simultaneous RTS/ CTS with different Originators (DUT as Originator)	J1939-21 5.10.5.1	Verify behavior (abort) with received multiple duplicate CTS (excluding CTS(0))	X										
16	Transport Data Packets for RTS/CTS have correct content (DUT as Originator)	J1939-21 5.10.1.3 5.10.2.3 5.10.4	Verify first data byte is sequence number Verify remaining 7 bytes are correct data for packet	X										

TABLE A7 - DATA LINK TRANSMIT TESTS - TP RTS/CTS (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
17	Transport Data Packets for RTS/CTS sent in ascending sequential order (DUT as Originator)	J1939-21 5.10.1.1 5.10.1.2 5.10.1.4	Verify DUT sends the TP Data Packets for an RTS/CTS in sequential order Verify the first TP.DT packet is sequence number = 1	X										
18	All Transport Data Packets for RTS/CTS have an 8 byte data field (DUT as Originator)	J1939-21 5.10.1.3 5.2.7.2 5.10.4	Verify DUT sends packets with 8 bytes	X										
19	Unused bytes of Last Transport Data Packets for RTS/CTS filled with "FF_{16}" (DUT as Originator)	J1939-21 5.10.1.3 5.10.4	Verify unused bytes of last packets sent & filled properly	X										
20	Transport Data Packets for RTS/CTS start within T2 time following CTS (DUT as Originator)	J1939-21 5.10.2.3 5.10.2.4	Verify packets are sent no greater than 200 mS apart	X										
21	Time between consecutive Transport Data Packets for RTS/CTS, in which the CTS that allows more than 1 packet, are sent within T2 time of one another (DUT as Originator)	J1939-21 5.10.1.3 5.10.2.4	Verify packets of a CTS set are sent no greater than 200 mS apart (says T1 of 750 mS in 5.10.2.4)	X										

A.3.6 Data Link Layer Tests - RTS/CTS Transport Protocol Responder Behaviors

The Compliance Tests in Table A8 evaluate the behaviors of the DUT as the data recipient (Responder) for the SAE J1939 Transport Protocol RTS/CTS service. The SAE J1939 Transport Protocol BAM tests for the data recipient are presented in Table A6.

TABLE A8 - DATA LINK RECEIVE TESTS - TP RTS/CTS

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
1	RTS/CTS Protocol: RTS is valid (content and format) (DUT as Responder)	J1939-21 5.10.2.2 5.10.3 5.10.3.1	Verify behavior when unsupported or unknown PGN Verify behavior for other reject reasons for the DUT	X										
2	RTS/CTS Protocol: Multiple duplicate RTS (same SA, DA, PGN) received in short period (DUT as Responder)	J1939-21 5.10.3.1	If DUT accepts, verify DUT only once (not to every RTS)	X										
3	RTS/CTS Protocol: CTS following RTS is valid (content and format) (DUT as Responder)	J1939-21 5.10.2.2 5.10.3 5.10.3.2	Verify CTS sent within 200 mS (Tr) following RTS Verify correct PGN, next packet(= 1), number of packets (<= RTS max packets)	X										
4	RTS/CTS Protocol: To reject, RTS is followed by valid Conn_Abort (content and format) (DUT as Responder)	J1939-21 5.10.2.2 5.10.3.4	Verify DUT Conn_Abort has correct PGN and a valid abort reason provided and is sent withing 200 mS (Tr) following RTS	X										
5	RTS/CTS Protocol: CTS issued after last Transport Data packet of the previous CTS (DUT as Responder)	J1939-21 5.10.2.2 5.10.2.4 5.10.3 5.10.3.2	Verify CTS sent within T3 time following last packet Verify correct PGN, next packet (valid), number of packets(<= RTS max packets)	X										

TABLE A8 - DATA LINK RECEIVE TESTS - TP RTS/CTS (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
6	RTS/CTS Protocol: CTS next packet numbering (DUT as Responder)	J1939-21 5.10.3.2	Verify next packet number specified in CTS is not less than packet number already confirmed Verify next packet number specified in CTS is either same as last CTS, within range of last CTS, or one greater than the next packet number plus number of packets from last CTS	X										
7	RTS/CTS Protocol: CTS issued after T2 time out during the transport of data packets (DUT as Responder)	J1939-21 5.10.3.2	Verify sends CTS (or Abort) after T2 time with no packets following CTS	X										
8	RTS/CTS Protocol: CTS to pause or stop data flow (DUT as Responder)	J1939-21 5.10.2.3 5.10.3 5.10.3.2	Verify correct setting of PGN and number of packets (0) and all other bytes set to 1's Verify CTS sent within T3 time following last packet	X										
9	RTS/CTS Protocol: CTS to pause repeat (DUT as Responder)	J1939-21 5.10.2.3 5.10.2.4 5.10.3 5.10.3.2	Verify correct setting of PGN and number of packets (0) and all other bytes set to 1's Verify CTS pause sent within 500 mS (Th) of last CTS pause Verify sent only after receiving all packets recently cleared (or expired receive time out)	X										

TABLE A8 - DATA LINK RECEIVE TESTS - TP RTS/CTS (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
10	RTS/CTS Protocol: EndOfMsgACK (DUT as Responder)	J1939-21 5.10.2.4 5.10.3 5.10.3.3	Verify correct content in TP.CM_EndOfMsgACK Verify sends a valid TP.CM_EndOfMsgACK only after the last data packet Verify sends a valid TP.CM_ EndOfMsgACK within T3 Time after last data packet (or CTS pause)	X										
11	RTS/CTS Protocol: Connect Abort issued by Originator (DUT as Responder)	J1939-21 5.10.2.4 5.10.3.4	Verify behavior when receives TP.Conn_Abort Verify ignores any Transport Data packets following Abort Verify DUT checks PGN	X										
12	RTS/CTS Protocol: Connect Abort issued by Responder (DUT as Responder)	J1939-21 5.10.2.4 5.10.3.4	Verify correct content in TP.Conn_Abort Verify ignores any Transport Data packets following Abort	X										
13	RTS/CTS Protocol: CTS outside of connection is ignored (DUT as Responder)	J1939-21 5.10.3.2	Verify doesn't randomly sent CTS for PGNs not in a connection	X										
14	RTS/CTS Protocol: Multiple duplicate CTS (same SA, DA, PGN) received in short period (DUT as Responder)	J1939-21 5.10.3.2	Verify doesn't send multiple duplicate CTS (excluding CTS(0))	X										

TABLE A8 - DATA LINK RECEIVE TESTS - TP RTS/CTS (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
15	RTS/CTS Protocol: Simultaneous RTS/ CTS with different Originators (DUT as Responder)	J1939-21 5.10.5.1	Verify ability of DUT to successfully manage Transport Data when RTS/CTS and BAM with same originator Verify DUT uses/evaluates Destination Address Verify doesn't send multiple duplicate CTS (excluding CTS(0))	X										
16	TP Protocol: Simultaneous RTS/CTS and BAMs with same Originator (DUT as Responder)	J1939-21 5.10.5.1	Verify ability of DUT to successfully manage Transport Data when RTS/CTS and BAM with same originator Verify DUT uses/evaluates Destination Address to distinguish TP.DT for RTS/CTS or BAM	X										
17	RTS/CTS Protocol: Another RTS with same Originator and Responder but with different PGN (DUT as Responder)	J1939-21 5.10.5	Verify behavior (DUT rejects by sending Conn_Abort) any additional RTS from an Originator while another RTS/CTS open with same Originator	X										
18	RTS/CTS Protocol: RTS received when ECU at TP Connection Limit (DUT as Responder)	J1939-21 5.10.5	Verify behavior (DUT rejects by sending Conn_Abort) any additional RTS while DUT at Connection Limit	X										
19	Transport Data Packets for RTS/CTS to be sent in ascending sequential order (DUT as Responder)	J1939-21 5.10.1.1 5.10.1.2 5.10.1.4	Verify DUT behavior when the TP Data packets for an RTS/CTS are received in order Verify behavior when TP Data packets for an RTS/CTS are received out of order	X										

TABLE A8 - DATA LINK RECEIVE TESTS - TP RTS/CTS (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
20	All Transport Data Packets for RTS/CTS have an 8 byte data field (DUT as Responder)	J1939-21 5.10.1.3 5.2.7.2 5.10.4	Verify behavior when receives packets with 8 bytes Verify behavior when receives packets with less than 8 bytes	X										
21	Transport Data Packets for RTS/CTS start within T2 time following CTS (DUT as Responder)	J1939-21 5.10.2.3 5.10.2.4	Verify behavior when packets are started within T2 time following CTS Verify behavior when packets are started after T2 time following CTS	X										
22	Time between consecutive Transport Data Packets for RTS/CTS, in which the CTS that allows more than 1 packet, are sent within T2 time of one another (DUT as Responder)	J1939-21 5.10.1.3 5.10.2.4	Verify behavior when packets are sent less than 200 mS apart (says T1 of 750 mS in 5.10.2.4) Verify behavior when packets are sent longer than 200 mS apart (says T1 of 750 mS in 5.10.2.4)	X										

A.4 TESTS OUTLINE FOR RP SAE J1939-31 RECOMMENDED PRACTICE FOR THE NETWORK LAYER

TABLE A9 - NETWORK LAYER TESTS

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
1	Address Claimed Message	J1939-31 4.4	Verify that the ECU does not go bus off due to the forwarding and contention of an Address Claimed message.		X									
2	Maximum messages forwarded	J1939-31 4.5	Verify that an ECU can forward the guaranteed maximum number of messages during average and peak busload.		X									
3	Maximum messages filtered	J1939-31 4.5	Verify that an ECU can forward and filter the guaranteed maximum number of messages during average and peak busload with the maximum number of entries in the database.		X									
4	Maximum transit delay	J1939-31 4.5	Verify that the worst-case latency for transmitting a message from one device to another device on a different bus segment does not exceed the maximum transit delay and is less than 50ms.		X									
5	Message Forwarding	J1939-31 5.1	Verify that messages can be forwarded from one segment to another.		X									
6	Higher Priority Forwarding	J1939-31 5.1	Verify that messages with higher priority are forwarded before messages with lower priority.		X									
7	Equal Priority Forwarding	J1939-31 5.1	Verify that messages with equal priority are forwarded in the order received.		X									
8	Message Filtering	J1939-31 5.2	Verify that messages can be filtered from one segment to another.		X									

TABLE A9 - NETWORK LAYER TESTS (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded										
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl	Date Tested	Pass-Fail	Test Result Comments
9	Block or Pass Filtering	J1939-31 5.2	Verify one of the following : 1.) Messages are forwarded by default, and blocked only if they exist in the filter database. 2.) Messages are blocked by default, and forwarded only if they exist in the filter database.		X									
10	Database management	J1939-31 5.6	Verify that entries added to the filtering databases can only be removed by the device that created the entry, or by diagnostic tools that have the ability to override the address match requirement.		X									
11	Address Translation	J1939-31 5.3	Verify that source/destination addresses can be substituted in messages, which use a single address to reference a particular vehicle system.		X									
12	Message Repackaging	J1939-31 5.4	Verify that messages can be repackaged and forwarded.		X									
13	Message Filter Database Manipulation	J1939-31 5.6.2	Verify that the filter database can be manipulated as follows: - Request entries - Add entries - Delete entries - Clear database - Set filter mode - Create entry, specifying filter mode		X									
14	General Parametrics	J1939-31 5.6.5	Verify that general parametrics can be accessed.		X									

TABLE A9 - NETWORK LAYER TESTS (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
15	Port Pair Parametrics	J1939-31 5.6.6	Verify that specific port pair parametrics can be accessed.		X									
16	Network Topology	J1939-31 5.6.7	Verify that network topology information can be accessed.		X									

A.5 TESTS OUTLINE FOR RP SAE J1939-81 RECOMMENDED PRACTICE FOR NETWORK MANAGEMENT

TABLE A10 - NETWORK MANAGEMENT TESTS

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
1	ECU NAME	J1939-81 4.1	Verify the NAME contents for the ECU (DUT) align with -81	X										
2	System Initialization	J1939-81 4.4.3	Verify each ECU transmits a address claim at power-up system initialization.	X										
3	Non-Configurable Address ECU	J1939-81 4.1.2 4.2.2.3	Verify that a non-configurable address ECU or service configurable address ECU stops transmitting and sends a Cannot Claim Address message if it fails to claim a valid address.	X										
4	Commanded Addresses	J1939-81 4.2.3	Verify that a command configurable address ECU can receive a Commanded Address message and either initiate an address claim procedure with the new address or, if it cannot claim the new address, issue an address claim for it's current address.				X							
5	Self-Configurable Address ECU	J1939-81 3.3.1.4 4.4.4	Verify that a self-configurable address ECU can re-calculate and claim another address if it is not successful in claiming the calculated address.					X						
6	Request for Address Claimed	J1939-81 4.2.1	Verify an ECU can send a request for address claimed and process responses.											

TABLE A10 - NETWORK MANANGEMENT TESTS (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
7	Request for Address Claimed	J1939-81 4.2.1	Verify an ECU responds to a request for address claimed with an Address Claimed/Cannot Claim message (or nothing if that ECU has not yet attempted to claim an address).	X										
8	Address Claimed Cannot Claim	J1939-81 4.2.2	Verify an ECU sends an Address Claimed message upon initialization and waits for the defined period (250 ms or 50 ms) before resuming normal network traffic.	X										
9	Address Claimed Cannot Claim	J1939-81 4.2.2	Verify an ECU receiving an Address Claimed message with a lower priority claiming it's own source address responds with an Address Claimed Message.	X										
10	Address Claimed Cannot Claim	J1939-81 4.2.2	Verify an ECU receiving an Address Claimed message with a higher priority claiming it's own source address either attempts to claim a new address or responds with a Cannot Claim message after a time delay.	X										
11	Address Not Claimed	J1939-81 4.2.2.3	Verify an ECU that cannot claim an address sends the Cannot Claim message in response to the Request for Address Claimed. No other messages shall be sent.	X										
12	Power Interruption	J1939-81 4.5.1	Verify an ECU does not perform network initialization with a power loss < 2ms.Verify an ECU does perform network initialization with a power loss > 1 second.		X	X	X	X		X				

TABLE A10 - NETWORK MANANEMENT TESTS (CONTINUED)

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
13	Network Disruption	J1939-81 4.5.2.3	Verify that an ECU can be connected, disconnected, and powered up without disrupting network communications.	X										
14	Address Continuity	J1939-81 4.5.2.4	Verify that an ECU attempts (if applicable) to use the same address and addresses for ECUs that are communicated with across power-down and power-up cycles.			X		X						

A.6 TESTS OUTLINE FOR RP SAE J1939-1X RECOMMENDED PRACTICE PHYSICAL LAYER

TABLE A11 - PHYSICAL LAYER TESTS

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
1	J1939-11 physical layer	J1939-11	Verify to requirements of SAE J1939-11 Conformance Tests.											
2	J1939-15 physical layer	J1939-15	Verify to requirements of SAE J1939-15 Conformance Tests.											

APPENDIX B - FORMAT FOR COMPLIANCE TEST REPORTING FORMS

The Tables herein shows the format that should be used for reporting compliance tests results. Such a form could serve for self-certifying compliance, as well as for ordering to identify what functions a given device can or must support.

B.1 COMPLIANCE TEST REPORTING FORMS

This form shows the format for reporting the results of compliance tests conducted to SAE J1939 Recommended Practice, Appendix A of SAE J1939-82 contains a set of forms listing all the compliance tests as well as this reporting formatting.

See SAE J1939-82 Appendix A Section A.2.5.2 for a definition of ‘Class’ for an ECU.

TABLE B1 - SAMPLE REPORTING FORM

Row	Test Name	Requiring Document	Description	SAE and User Requirements X: Test is Recommended D: Additional Test Desired E: Test Excluded								Date Tested	Pass-Fail	Test Result Comments
				ECU Classes							User			
				ALL	BDG	AAC	CC	SC	TOO	EMS	Add./ Excl			
1														
2														
3														
4														
5														
6														